INFS1200/7900 Module 2 Assignment

Due: 18 September 2020 @ 11:59 PM AEST

Weighting: 10%

	Full Name	Student ID
Group Member 1		
Group Member 2		

1. Overview

The purpose of this assignment is to test your ability to apply relational model principles, such as identifying integrity constraint violations in common database operations and mapping from ER to relational schema.

This assignment can be completed either in groups of two or individually.

2. Submission

All submissions must be made through an electronic marking tool called Gradescope, which will also be used for providing feedback. You **must** record all your answers in the spaces provided in this document. Altering the format or layout of this document in anyway will attract penalties.

3. Marking

The Module 2 assignment is worth 10 course marks (of 40 course marks total for all four assignments).

4. Plagiarism

The University has strict policies regarding plagiarism. Penalties for engaging in unacceptable behaviour can range from cash fines or loss of grades in a course, through to expulsion from UQ. You are required to read and understand the policies on academic integrity and plagiarism in the course profile (Section 6.1).

If you have any questions regarding acceptable level of collaboration with your peers, please see either the lecturer or your tutor for guidance. Remember that ignorance is not a defence!

5. Task

This assignment contains two different sections. Section A will contain a relational model schema along with other contextual information. Based on this information, you will need to analyse ten different database operations. For each, identify which, if any, integrity constraints would be violated by these operations and how they could be remedied by the database system. An example has been provided below.

Section B will provide you with three ER diagrams based off sample solutions from Assignment 1. For each of these diagrams, you will need to perform relational mapping and state the final relational schema for the ER diagram, including any foreign keys which were created during the process.

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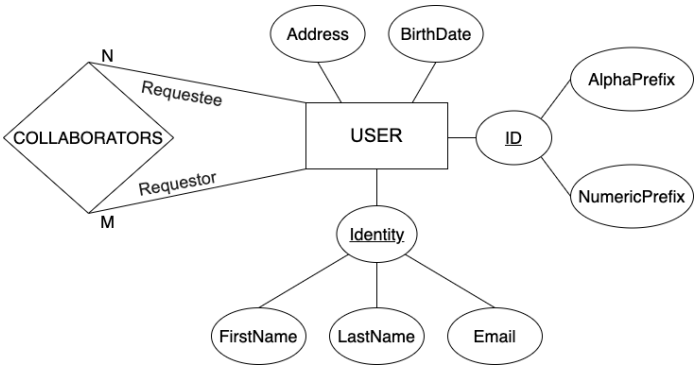
Section A – Integrity Constraints

The following is an extract from a collaboration platform relational system. The original ER diagram has been provided to highlight additional key constraints which are not intuitive from the relational schema. Using the ER diagram, contextual information, relational schema and instance data provided, answer the following questions. **Note: You do not need to take into consideration changes which may have been made by operations in earlier questions**.

ER Diagram:



Contextual Information:

In order to promote users developing more meaningful collaborations with other users, no user is allowed to request to collaborate with more than two users at any given time. Additionally, if a user has only one collaborator, that requestee is not allowed to stop collaborating with that user.

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Relational Schema:

User [alphaPrefix, numericSuffix, firstName, lastName, email, address, birthDate]

Collaborators [requestorAlphaPrefix, requestorNumericSuffix, requesteeAlphaPrefix, requesteeNumericSuffix]

Foreign Keys:

Collaborators.{requestorAlphaPrefix, requestorNumericSuffix} references
User.{alphaPrefix, numericSuffix}

Collaborators.{requesteeAlphaPrefix, requesteeNumericSuffix} references
User.{alphaPrefix, numericSuffix}

Instance Data:

User

alphaPrefix	numericSuffix	firstName	lastName	email	address	birthDate
ABC	123456	Bill	Mush	bill@gmail.com	7 Right Lne	20/04/1996
QUR	358732	Ellen	Lee	lee@outlook.edu	2 Alpha Dr	12/11/1987
RBT	987652	Frank	Bardon	bar@brisbane.com	1/1 Main St	08/04/1998
ABC	164578	Grace	Smith	soccer@bris.bane	45 Main St	13/06/1990
XZY	286843	Aaron	Smith	soccer@bris.bane	46 Main St	14/07/1991

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Collaborators

requestorAlphaPrefix	requestorNumericSuffix	requesteeAlphaPrefix	requesteeNumericSuffix
ABC	123456	QUR	358732
QUR	358732	ABC	123456
ABC	123456	ABC	164578
ABC	164578	RBT	987652
XZY	286843	ABC	164578

Example: Insert (125395, 097123, 'Jian', 'Zhang', 'j.zhang@uq.edu.au', '46 University Dr', '12/12/1995') into User.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated: Domain Constraint Violation

If yes, explain how these integrity constraints could be enforced:

This constraint could be enforced by rejecting the insert operation into the database. Additionally, the DBMS could replace the incorrect 'alphaPrefix' with a default value provided it does not violate the key constraint.

A.1 Insert (NULL, 097123, 'Grace', 'Zhang', 'bubbletea@uq.edu.au', '40 University Dr', '12/12/1995') into User.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

A.2 Insert ('RBT', 987652, 'RBT', 987652) into Collaborators.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

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A.3 Insert ('ABC', 987652, 'RBT', 987652) into Collaborators.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

A.4 Modify the tuple ('ABC', 123456, 'ABC', 164578) in the Collaborators table and change it to ('ABC', 123456, 'RBT', 987652).

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

A.5 Modify the 'requestorNumericSuffix' for all the tuples in Collaborators table with 'requestorAlphaPrefex' = 'ABC' to 123456.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

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A.6 Modify the 'firstName' of the User tuple with 'alphaPrefix' = 'XZY' and 'numericSuffix' = 286843 to 'Grace'.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

A.7 Delete any tuple in Collaborators with 'requestorNumericSuffix' = 'QUR'.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

A.8 Delete any tuple in the Collaborators table with 'requestorNumericSuffix' = 164578.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

A.9 Delete the tuple ('ABC', 123456, 'Bill', 'Mush', 'bill@gmail.com', '7 Right Lne', '20/04/1996') in the User table.

Will this operation cause an integrity constraint violation? Yes / No

If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

A.10 Insert the tuple ('XZY', '097123', 'Bob', 'Smith', 'infs1200@uq.edu.au', '48 University Dr', '12/11/1998') into User.

Will this operation cause an integrity constraint violation? Yes / No

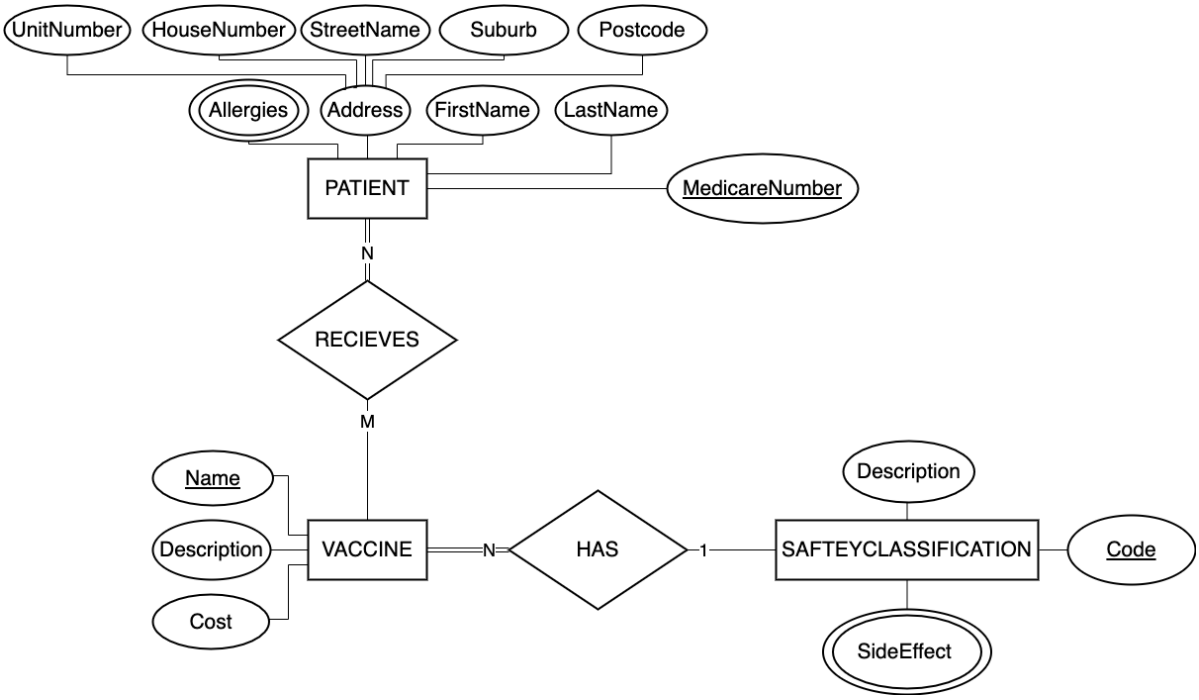
If yes, name the constraint(s) which will be violated:

If yes, explain how these integrity constraints could be enforced:

Section B – Relational Mapping

The following three questions all contain ER or EER diagrams from the Module 1 assignment solutions. For each diagram, complete ER to relation mapping and state your final relational schema, including any foreign keys which were created in the process.

B.1



ENTER YOUR RELATIONAL SCHEMA AND FOREIGN KEYS HERE

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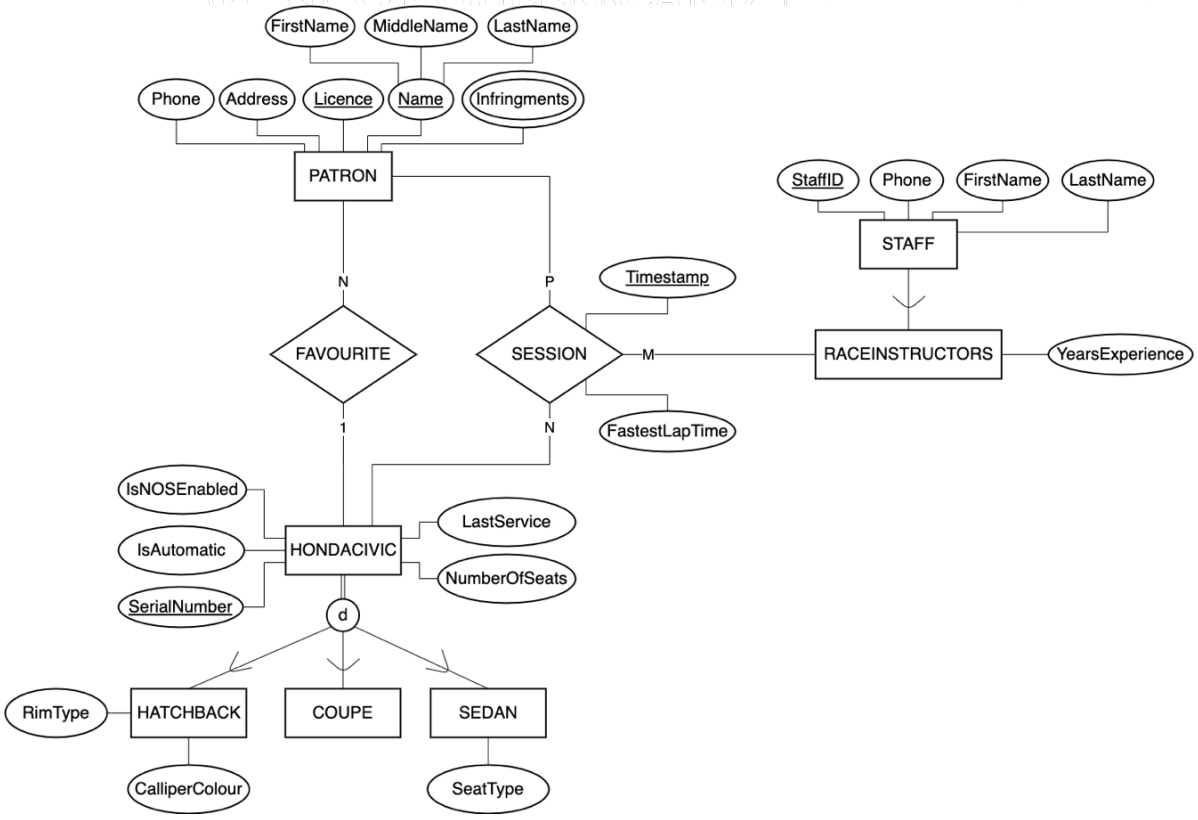
B.2



ENTER YOUR RELATIONAL SCHEMA AND FOREIGN KEYS HERE

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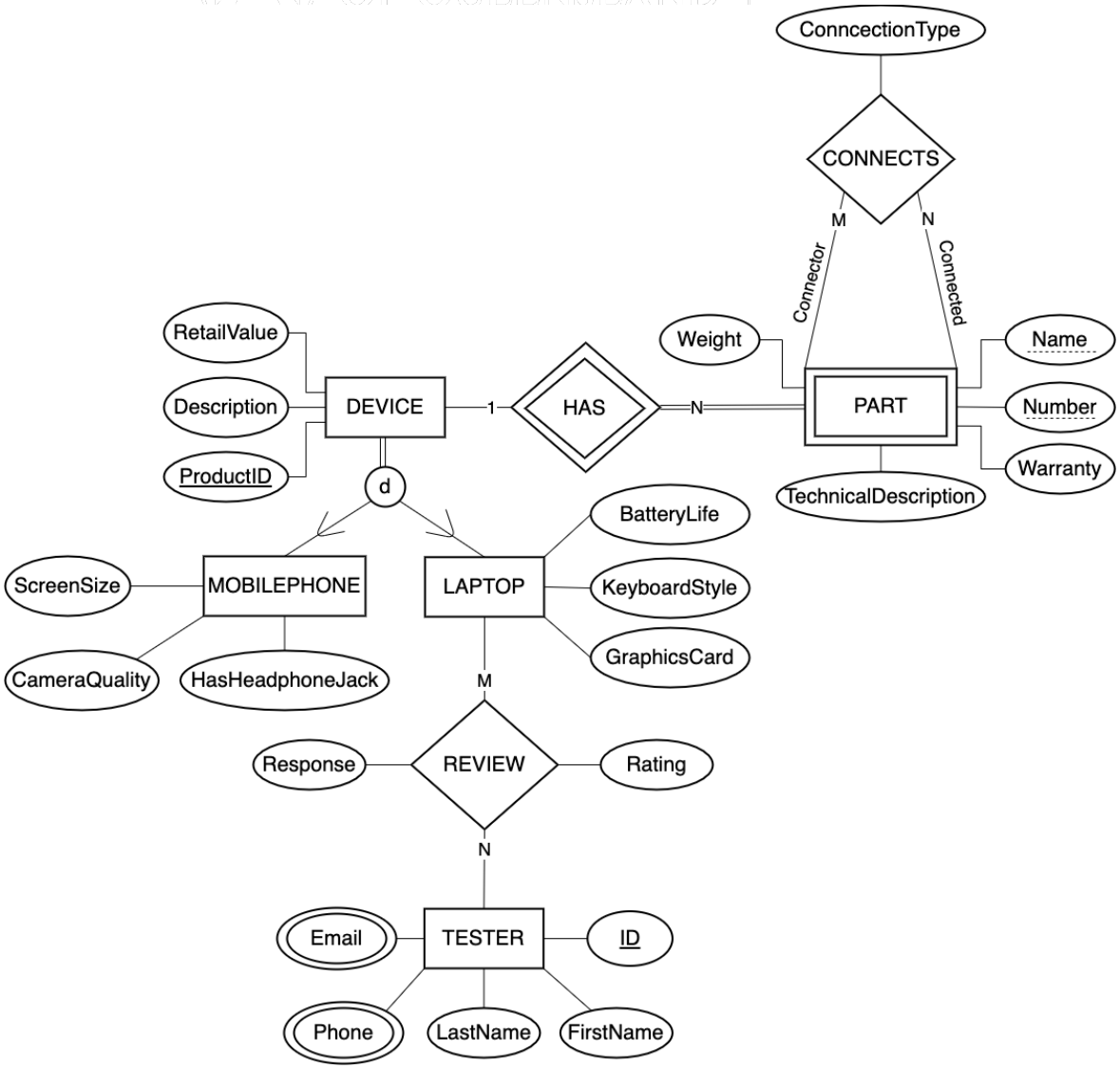
B.3



ENTER YOUR RELATIONAL SCHEMA AND FOREIGN KEYS HERE